Steering Protein Generative Models at Test-Time for Guided AAV2 Capsid Design

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Overview

- Recent advances in **protein generative models** have created new opportunities for protein engineering.
- However, a significant challenge remains in **effectively steering these models** to **generate sequences** with **specific, desired functionalities**, especially when these properties are defined by "black-box" or non-differentiable fitness functions.
- We introduce ProVADA+, a **model-agnostic framework** that **guides pretrained generative models** at test-time.
- We apply ProVADA+ to design **AAV2 capsids** with enhanced *in silico* capsid viability score.

Key Contributions

Likelihood-free, implicit priors

ProVADA+ is compatible with any base generative model, without requiring explicit density or likelihood evaluations.

Ensemble-guided test-time steering

ProVADA+ optimizes a composite functional objective to leverage an **ensemble of expert models**, **steering sequence generation in a fully gradient-free** manner.

Mixture-Adaptation Directed Annealing (MADA)

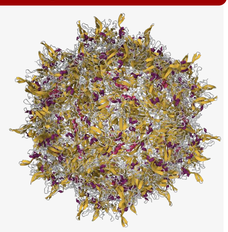
Developed a **novel sampling framework** combining population annealing with directed local mutations, enabling **highly efficient exploration of complex fitness landscapes**.

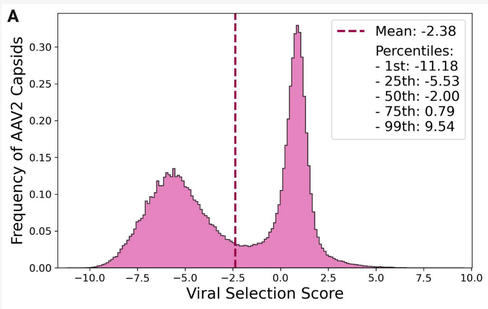
Adaptive position selection to mutate

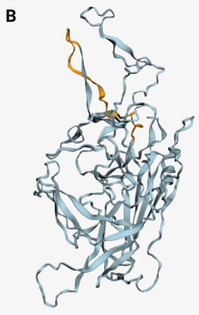
Introduced a novel strategy to construct an **informed proposal distribution for selecting mutation sites** based on **reinforcement learning**.

Adeno-associated Viruses (AAV)

- AAVs are a leading platform for *in vivo* gene therapy vector development
- For our experiments, we utilized the AAV2 capsid viability dataset from Bryant et al. which experimentally characterized 293,574 AAV2 variants for capsid viability.







Composite Fitness Objective

Sequence divergence

$$d_{\mathrm{Ham}}(x_0, x) = \sum_{\ell=1}^{L} \mathbb{1}\{x_\ell \neq x_{0,\ell}\},$$

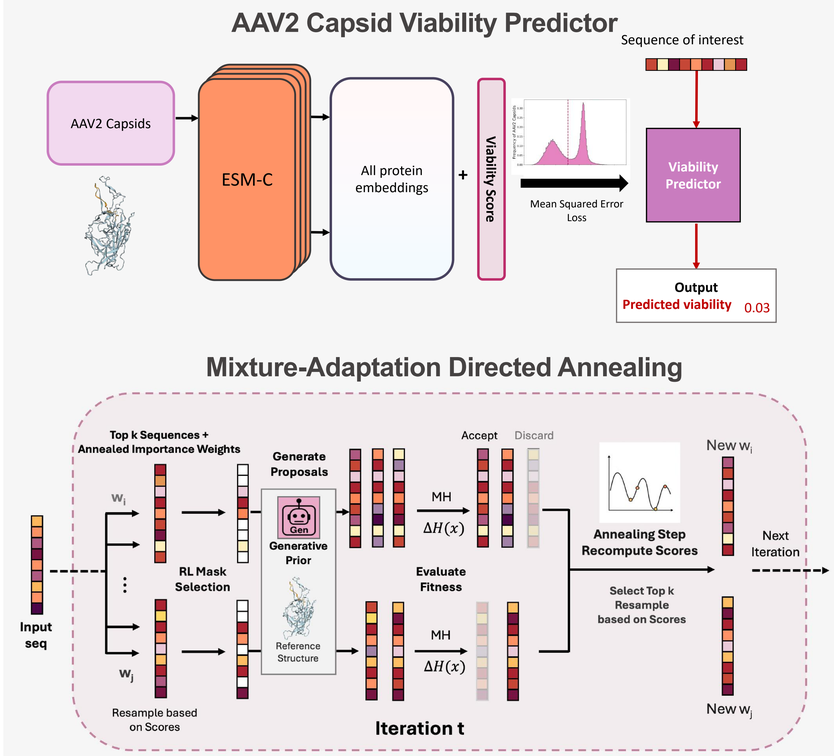
Adjusted fitness

$$H_\lambda(x) = \underbrace{[F(x) - \lambda\, D(x_0, x)]}_{\text{adjusted fitness}},$$

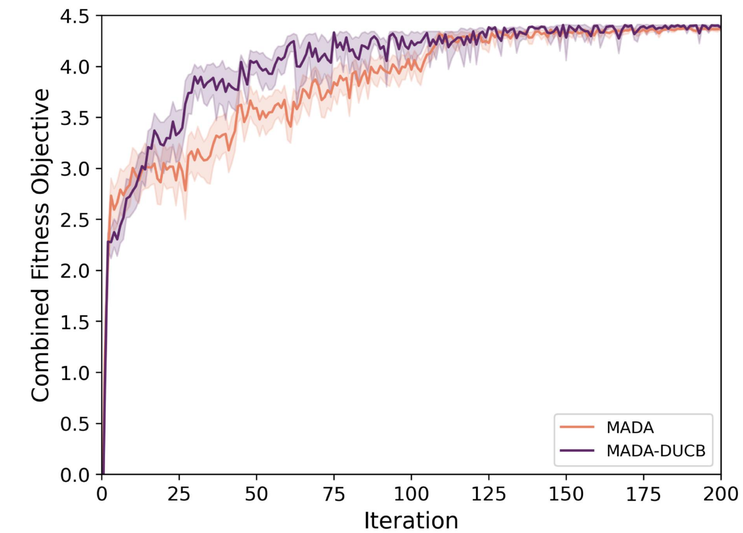
Gibbs posterior

$$\pi_{\phi,\tau,\lambda}(x) = \frac{p_\phi(x)\exp\big(H_\lambda(x)/\tau\big)}{\int p_\phi(x')\exp\big(H_\lambda(x')/\tau\big)\,dx'},$$

Test-Time Steering with ProVADA+

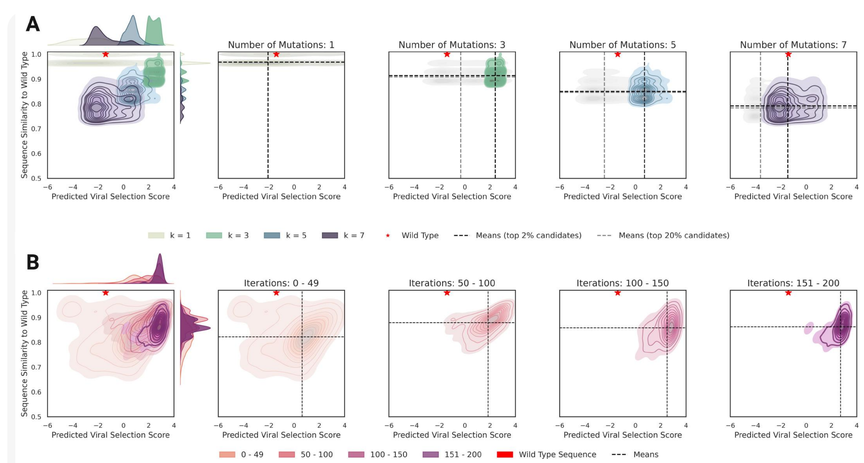


Faster convergence with ProVADA+



ProVADA+ with informed masking strategy (MADA-DUCB) shows improved convergence.

ProVADA+ Significantly Outperforms Current Methods



Joint distributions of predicted viral selection score and similarity to wild-type of the sequences generated through A) In silico selected variants from random mutagenesis; B) ProVADA-guided design. Variants generated via ProVADA+ consistently exhibit high viral selection scores while maintaining adequate diversity.

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